HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and implemented **preventive maintenance strategies** for critical fill and finish equipment, increasing overall equipment effectiveness by **12%** over a six-month period.
- Conducted **root cause analysis (RCA)** and **failure mode effects analysis (FMEA)** on recurring equipment failures, identifying design flaws in ancillary units that led to **45% fewer** unplanned stoppages.
- Monitored and analyzed equipment performance metrics including **MTBF** and **MTTR** for a fleet of 50+ lyophilizers and filling stations, providing actionable insights to optimize maintenance schedules.
- Provided **technical expertise** and troubleshooting support for complex issues on **aseptic processing** equipment, collaborating with production teams to resolve constraints and maintain cGMP compliance.
- Supported **qualification and validation activities** for equipment modifications, contributing to the successful integration of automated inspection systems into the sterile manufacturing workflow.
- Collected, analyzed, and reported on **reliability metrics and KPIs** using statistical methods, identifying trends that informed proactive intervention plans and reduced emergency repairs by **20%**.
- Managed change control processes for equipment upgrades and modifications, ensuring all documentation adhered to **cGMP, FDA, and EMA** regulatory requirements.

SKILLS

Reliability Engineering – RCA, FMEA, RCM, MTBF, MTTR, OEE, Predictive Maintenance, Preventive Maintenance, Condition-Based Monitoring

Manufacturing & Operations – Fill and Finish, Sterile Manufacturing, Aseptic Processing, GMP, FDA, EMA

Data Analysis & Tools – Statistical Analysis, Data Visualization, Microsoft Office Suite, CMMS, Python

Engineering Principles – Mechanical Engineering, Industrial Engineering, Electrical Engineering

Methodologies – Lean, Six Sigma, Project Management

PROJECTS

Fill Finish Equipment Reliability Optimization

- Analyzed historical maintenance logs and failure data for a portfolio of filling and lyophilization equipment to identify chronic issues and areas for improvement.
- Developed and proposed optimized **preventive maintenance schedules** based on equipment criticality and failure history, projecting a **15% reduction** in unscheduled downtime.
- Created **data visualization dashboards** in a BI tool to track key reliability metrics (MTBF, MTTR, availability) for production equipment, enabling faster identification of performance degradation.

Aseptic Processing Yield Improvement Initiative

- Investigated equipment-related factors impacting yield in aseptic processing lines, focusing on filling and sealing stations.
- Performed **FMEA** on critical process steps to identify potential failure modes and their impact on product sterility and throughput.
- Collaborated with cross-functional teams to implement corrective actions, contributing to a **7% increase** in overall line yield within one quarter.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)